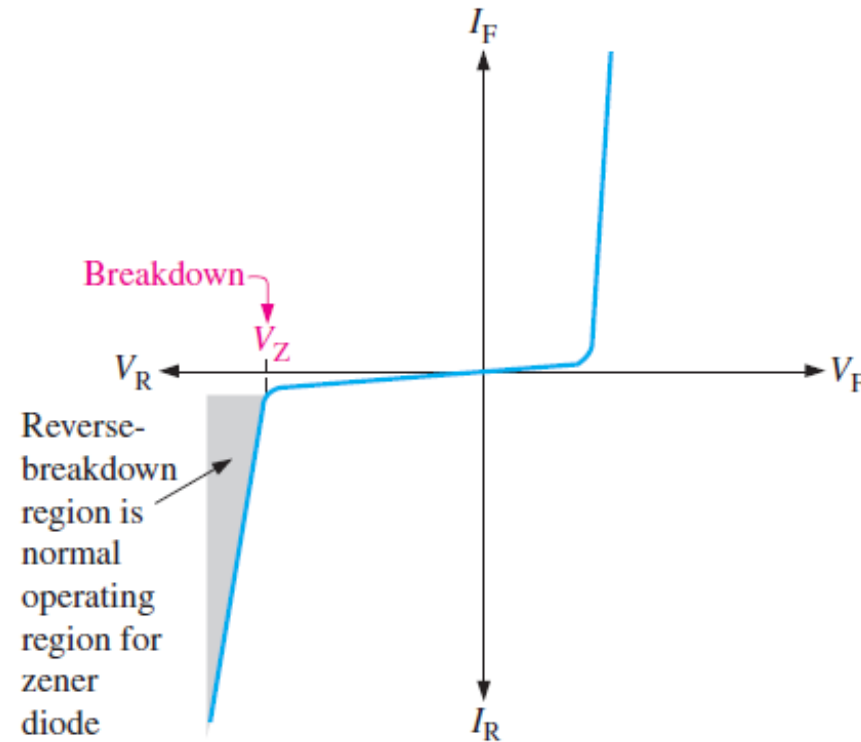
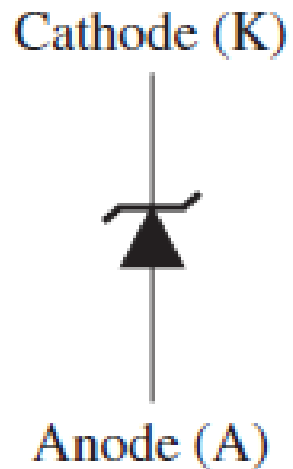


Zener Diode

- Zener diode is a typical diode which is designed to operate in reverse-breakdown region



Zener Diode (contd.)

- Two types of reverse breakdown in Zener diodes are observed i.e. avalanche and Zener
- Avalanche breakdown occurs at higher voltage levels but Zener breakdown occurs at low voltages
- Zener diode is heavily doped to reduce the breakdown voltage
- An intense electric field is generated in depletion region
- When applied voltage is near Zener breakdown voltage, the field is intense enough to pull the electrons from valence band to conduction band

Avalanche and Zener Effect

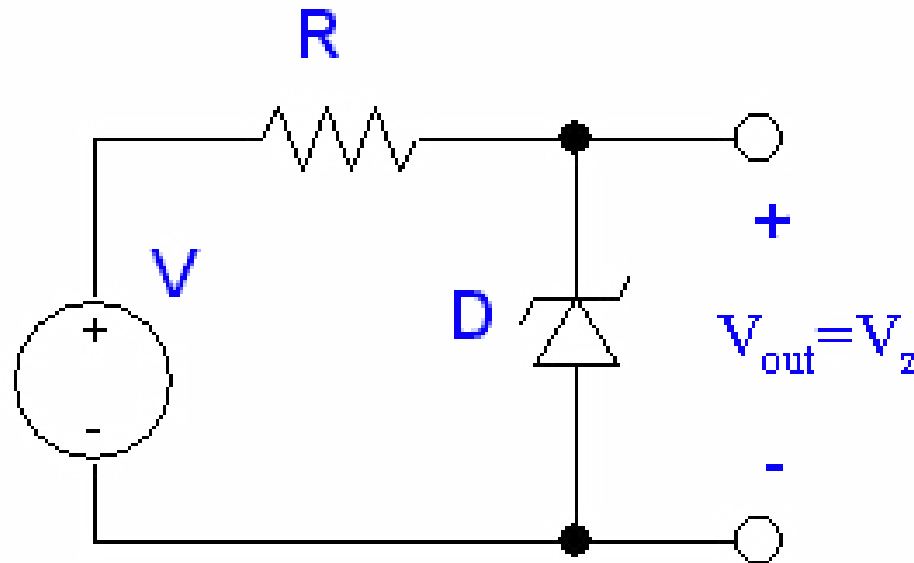
- Avalanche effect is observed when the material is lightly doped
- Zener effect is observed when material is heavily doped
- Width of the depletion layer is depend on the amount of doping
- Heavily doped diodes has narrow depletion layer and lightly doped diodes has wider depletion layer

Avalanche and Zener Effect (contd.)

- In reverse bias, a small reverse current is observed due to minority carriers
- When the applied voltage increases, it accelerate those minority carriers
- Those minority carriers then collide with majority carriers and knock them out
- This knocking out effect continues and hence current start to flow because of those majority carriers

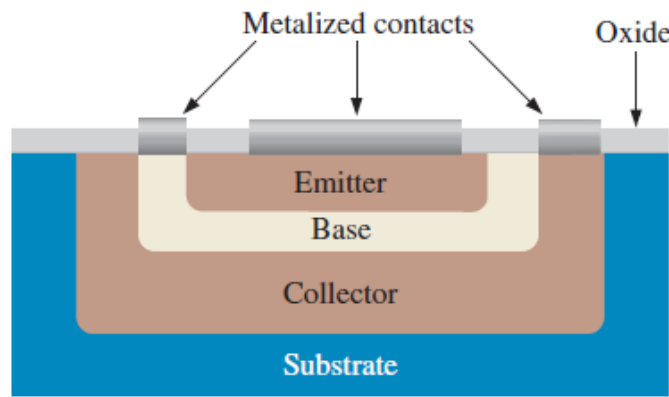
Zener Diode (contd.)

- Main applications of Zener diode are voltage regulators
- It can be used where a constant voltage is required (without fluctuations)

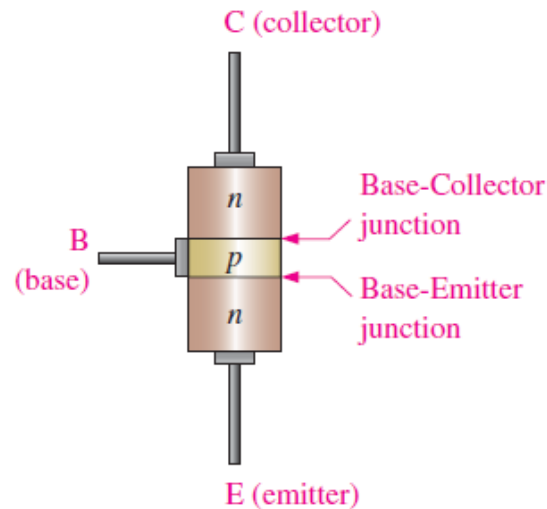


Bipolar Junction Transistors (BJT)

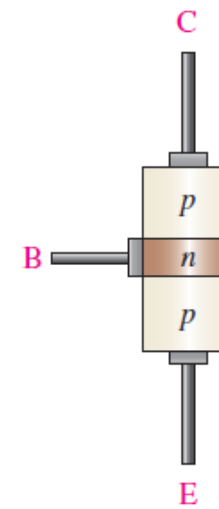
- BJT is constructed when three different semiconductor regions are joined together
- Three semiconductor regions are separated by two pn junctions
- Three regions are called Emitter, Base and Collector



(a) Basic epitaxial planar structure



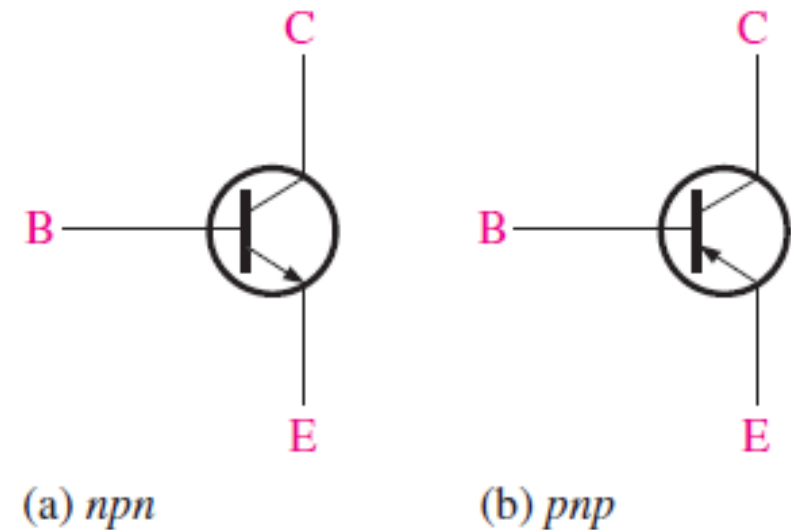
(b) npn



(c) pnp

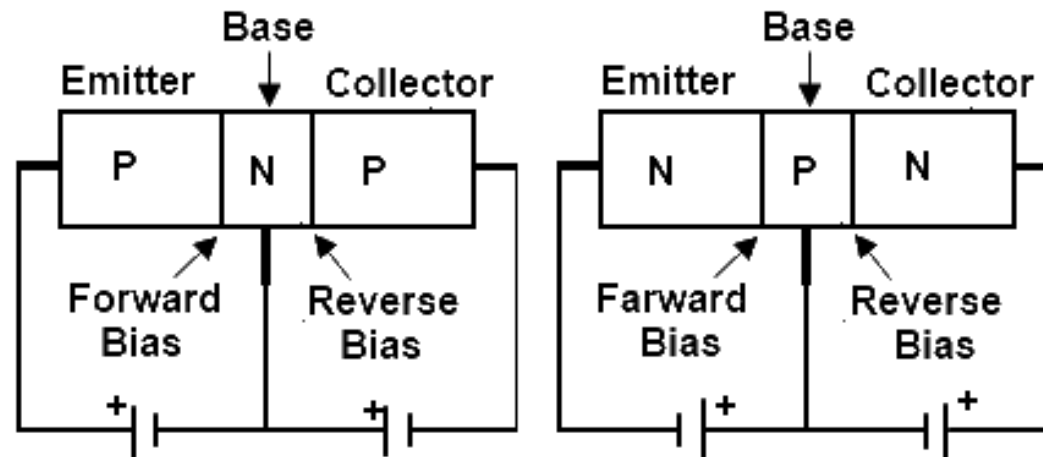
Bipolar Junction Transistors (contd.)

- The pn junction joining the base region and the emitter region is called Base Emitter junction (Emitter Diode)
- The pn junction joining the base region and collector region is called Base Collector junction (Collector Diode)
- Base region is lightly doped and very thin
- Emitter is heavily doped and collector is moderately doped
- Schematic symbol of BJT is shown in fig



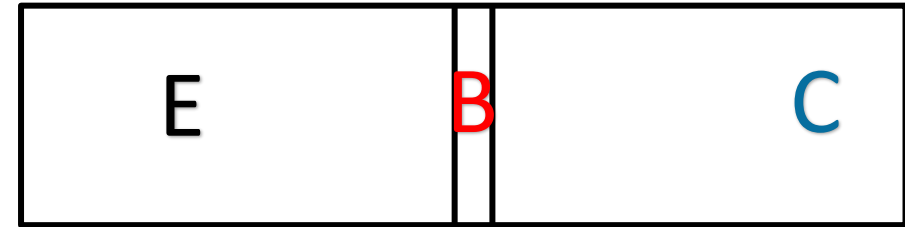
Bipolar Junction Transistors (contd.)

- In normal configuration/operation Emitter diode is forward biased and Collector diode is reverse biased
- Emitter has a job to emit its electrons so that they can inject in the base region
- When emitter diode is forward biased, electrons can enter from emitter to base



Bipolar Junction Transistors (contd.)

- Because of biasing, the electrons which enter in base has two options

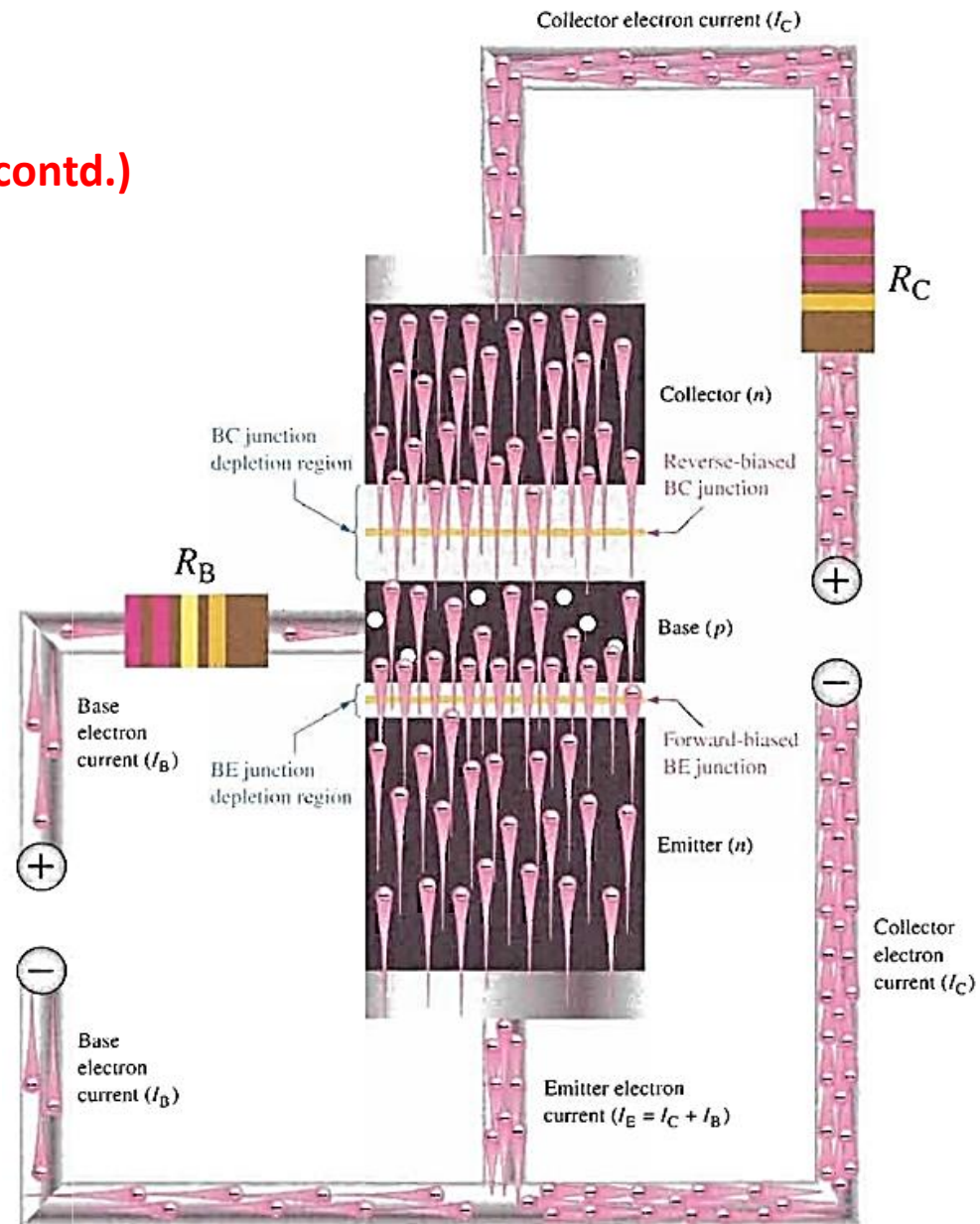


- To enter to the collector OR
 - To go out from the base
- Majority of the electrons will enter the collector as base is lightly doped and very thin
- Lightly doped means electrons have longer life in base and because of very thin base electrons have to move very short distance to enter into collector

Bipolar Junction Transistors (contd.)

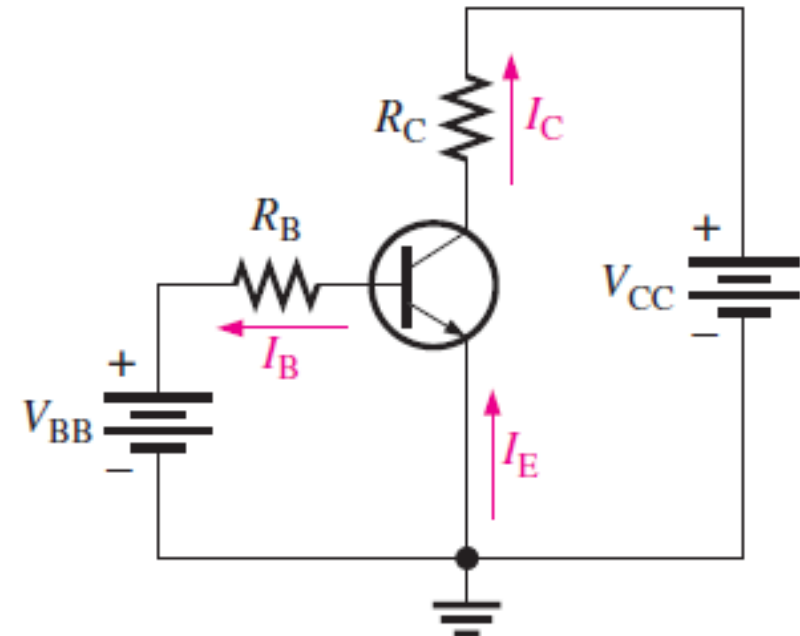
- When electrons enter into the collector, they feel a strong attraction because of the source voltage
- Because of this electrons flow through the collector and reach to the positive terminal of the source
- There are three useful configurations of transistors
 - Common Emitter
 - Common Base and
 - Common Collector

Bipolar Junction Transistors (contd.)



Common Emitter

- In this type of configuration, ground side or common side of each voltage source is connected to the emitter
- Hence it is called common emitter circuit/configuration
- The circuit has two loops which are
 - Left loop is called the base loop and
 - Right loop is called the collector loop



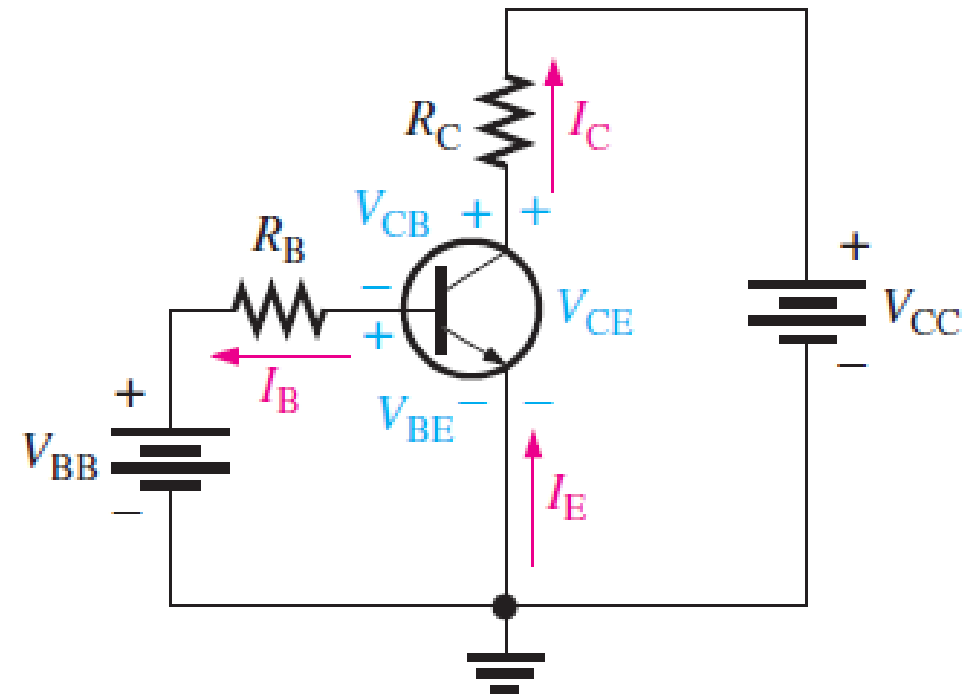
Common Emitter (contd.)

- In the base loop, the voltage source V_{BB} forward bias the emitter diode with R_B which is current limiting resistor
- If we change the V_{BB} or R_B , it means we can change the current
- Changing the base current will change the collector current it means base current will control the collector current

BJT Circuit Analysis

- From the circuit diagram, three different dc currents and three dc voltages can be identified
 - DC currents are I_B , I_E and I_C
 - DC voltages are V_{BE} , V_{CB} and V_{CE}
- When a transistor is in normal operation then

$$V_{BE} \cong 0.7V$$



BJT Circuit Analysis (contd.)

- By using Kirchof's voltage law

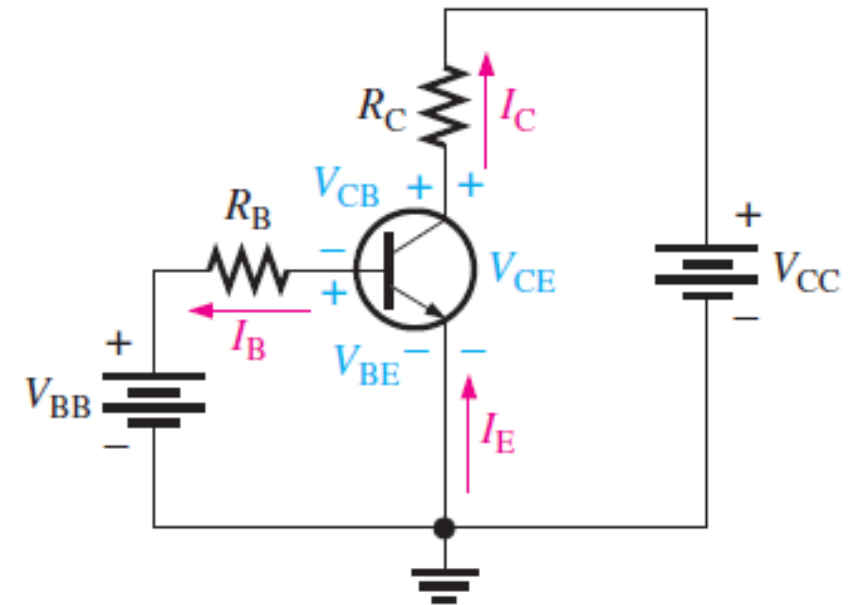
$$V_{RB} = V_{BB} - V_{BE}$$

- By Ohm's Law

$$V_{RB} = I_B R_B$$

- By substituting the values

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$



BJT Circuit Analysis (contd.)

- The voltage at collector can be calculated as:

$$V_{CE} = V_{CC} - V_{RC}$$

- Drop across the collector resistor is

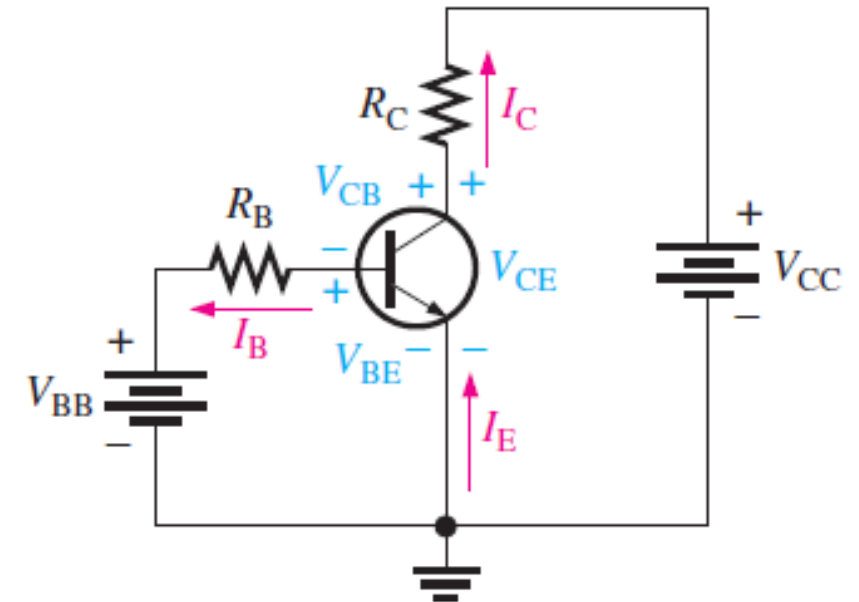
$$V_{RC} = I_C R_C$$

- By substituting the values

$$V_{CE} = V_{CC} - I_C R_C$$

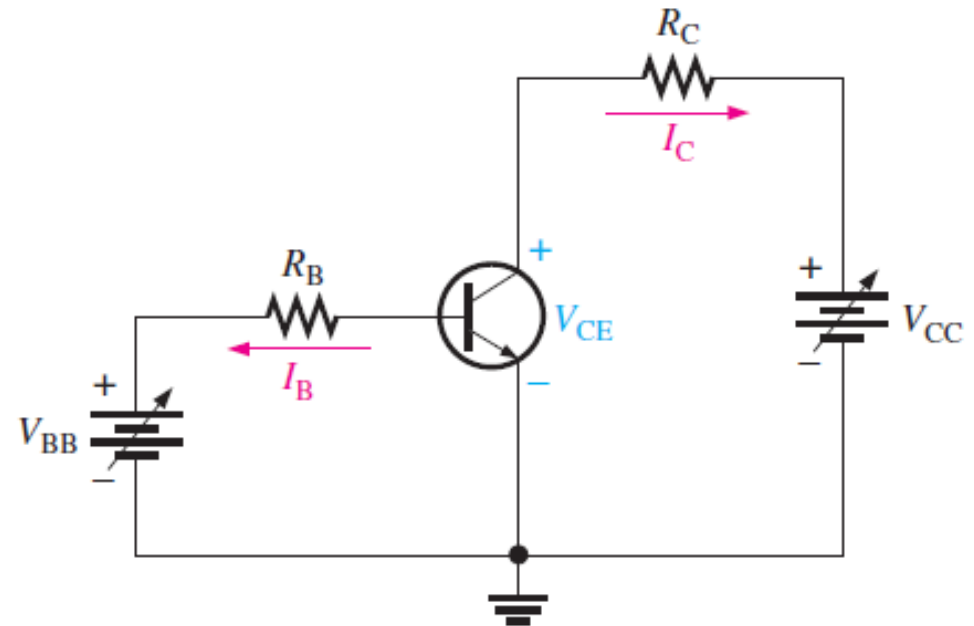
- Where

$$I_C = \beta_{DC} I_B$$



Collector Characteristic Curves

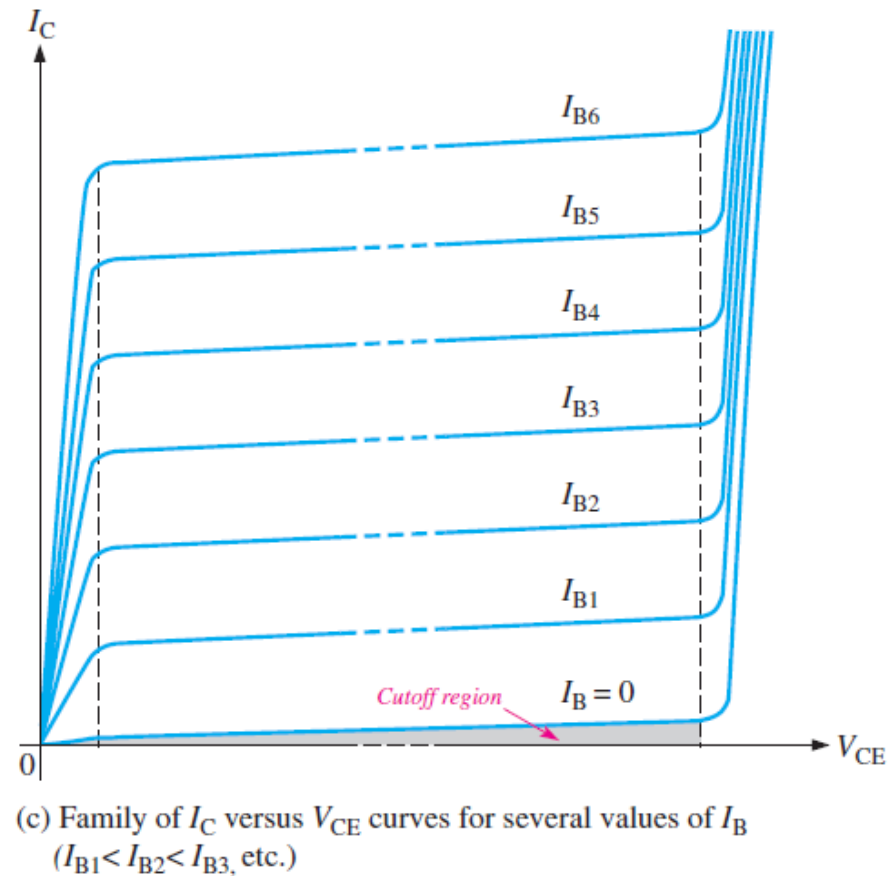
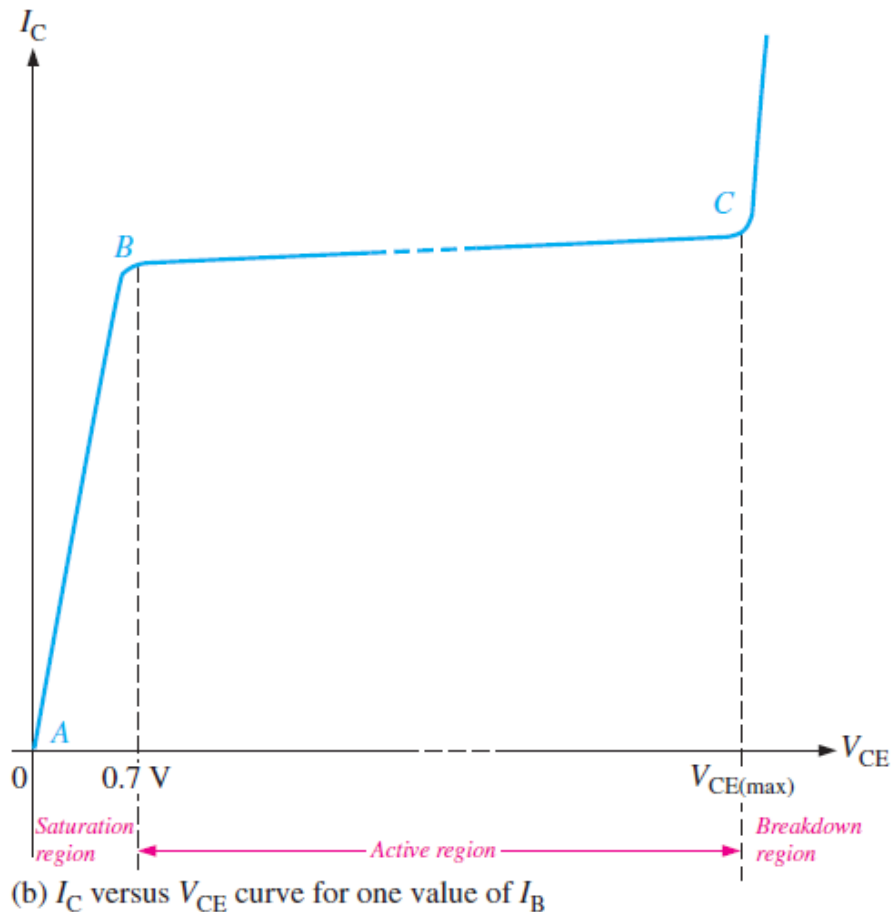
- Collector characteristics curves shows us that how the collector current varies the collector to emitter voltage for specific values of base current
- Assume that V_{BB} is producing a certain values of I_B and V_{CC} is zero
- Current is flowing only through base loop and I_C is zero
- As V_{CC} increases, V_{CE} increases and collector currents increases



Collector Characteristic Curves (contd.)

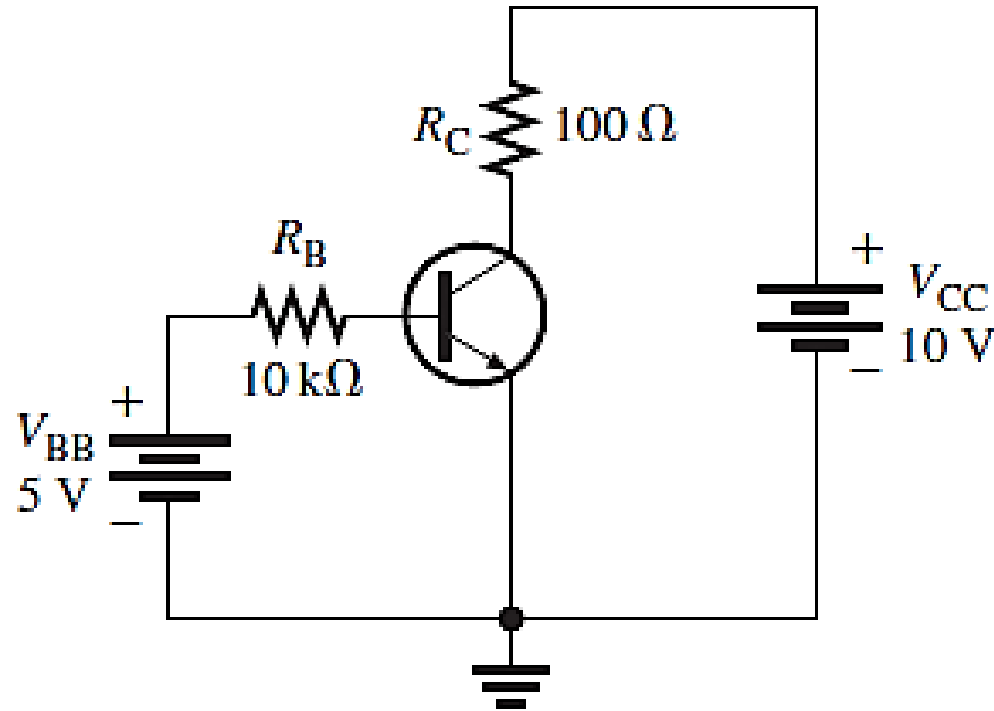
- This region is known as saturation region, I_C increases as V_{CC} increases because V_{CE} remains less than $0.7V$
- When V_{CE} exceeds $0.7V$, the collector diode becomes reverse biased and transistor goes into active region or linear region
- Ideally I_C remains constant as V_{CE} increases but I_C increases very slightly because of mobility of minority carriers
- When V_{CE} is large enough, the reverse biased collector diode goes into the breakdown region and I_C increases rapidly
- When I_B is zero, then transistor is in cutoff region

Collector Characteristic Curves (contd.)



A simple Circuit Analysis

- Calculate the value of I_B , I_C , I_E , V_{BE} , V_{CE} and V_{CB} for the circuit shown where $\beta_{dc} = 150$.



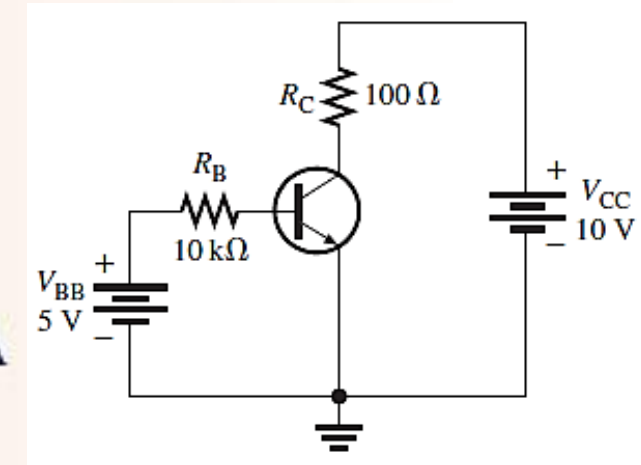
A simple Circuit Analysis (Solution)

~~From Equation 4-3~~ $V_{BE} \cong 0.7 \text{ V}$. Calculate the base, collector, and emitter currents as follows:

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{5 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = 430 \mu\text{A}$$

$$I_C = \beta_{DC} I_B = (150)(430 \mu\text{A}) = 64.5 \text{ mA}$$

$$I_E = I_C + I_B = 64.5 \text{ mA} + 430 \mu\text{A} = 64.9 \text{ mA}$$



Solve for V_{CE} and V_{CB} .

$$V_{CE} = V_{CC} - I_C R_C = 10 \text{ V} - (64.5 \text{ mA})(100 \Omega) = 10 \text{ V} - 6.45 \text{ V} = 3.55 \text{ V}$$

$$V_{CB} = V_{CE} - V_{BE} = 3.55 \text{ V} - 0.7 \text{ V} = 2.85 \text{ V}$$

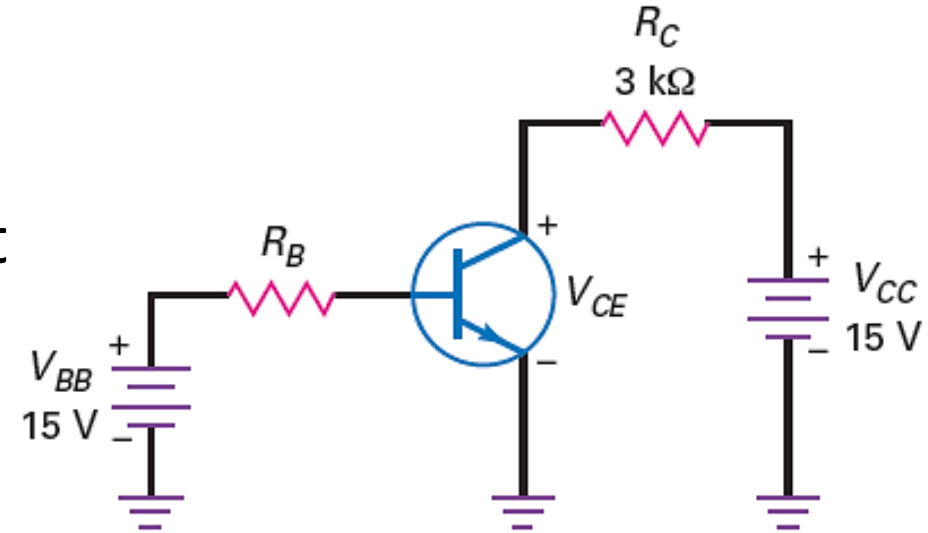
Since the collector is at a higher voltage than the base, the collector-base junction is reverse-biased.

Q Point of BJT

- Q point is also called Quiescent Point of transistor or DC operating point of transistor
- We have base bias circuit, means a fix base current is flowing
- If R_B has values of 1Mohm then base current will be $14.3\mu A$ and if β_{dc} has value of 100 then collector current will be equal to **1.43mA** and

$$V_{CE} = V_{CC} - I_c R_c$$
$$V_{CE} = 15 - (1.43 \times 10^{-3})(3 \times 10^3)$$
$$\mathbf{V_{CE} = 10.7V}$$

Therefore the **Q point** of the transistor will be **(10.7V, 1.43mA)**



Graphical Solution

- Q point can also be find out by graphically
- We know that

$$V_{CE} = V_{CC} - I_c R_c$$

$$I_c = \frac{V_{CC} - V_{CE}}{R_c}$$

- If we plot this equation (I_c versus V_{CE}) we will get a straight line and that line is called **Load Line** (if represent the effect of load on transistor)
- The end of this line can be easily found, first put V_{CE} equal to zero and then I_c equal to zero

Graphical Solution (contd.)

- When V_{CE} equal to zero, we get

$$I_c = \frac{15V - 0}{3k\Omega} = 5mA$$

- So first end point will be (0V, 5mA)

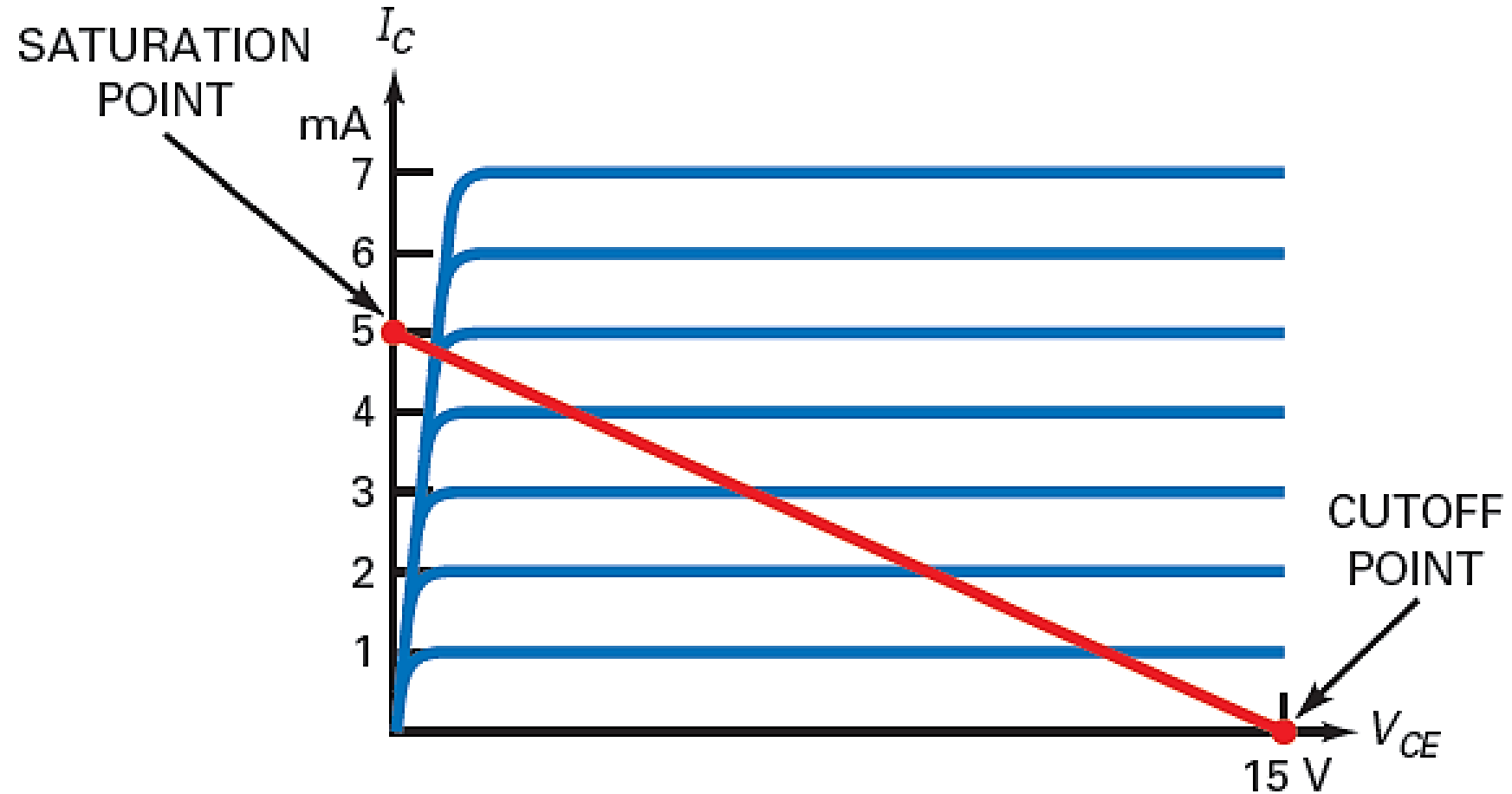
- When I_c equal to zero, we get

$$0 = \frac{15V - V_{CE}}{3k\Omega} \Rightarrow V_{CE} = 15V$$

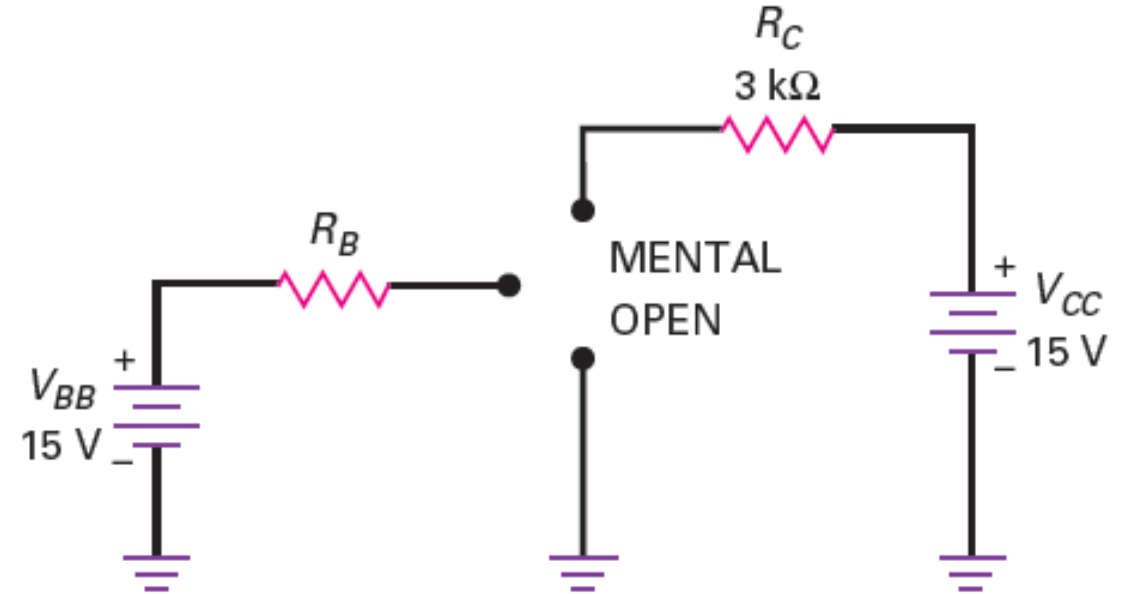
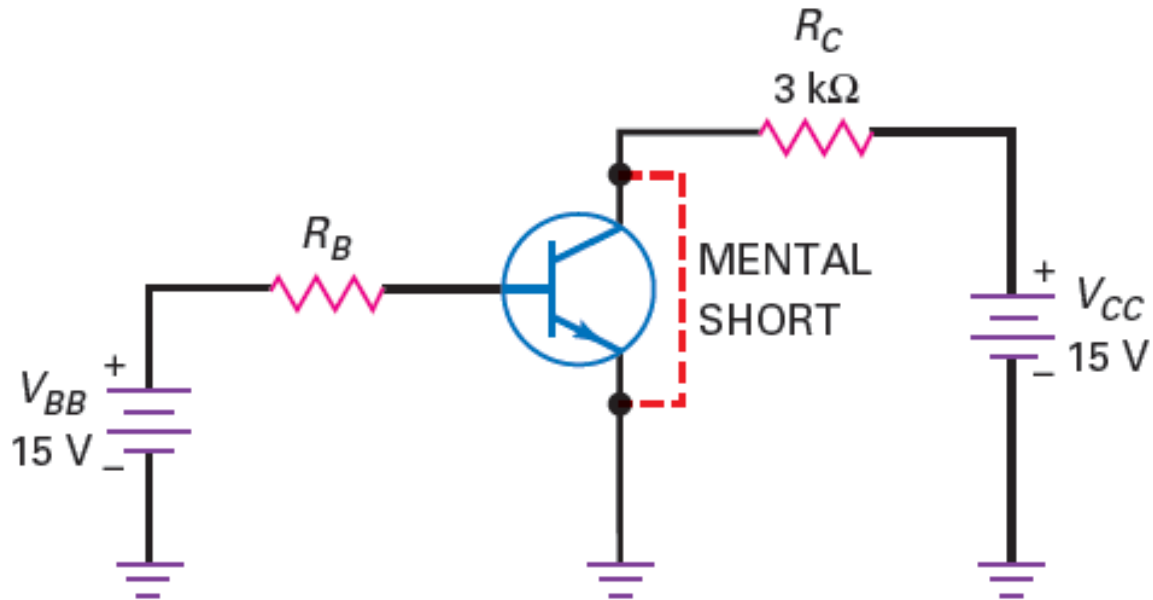
- And second end point will be (15V, 0Amp)

- By plotting the above points we will get a straight line called **Load Line**

Graphical Solution (contd.)



Graphical Solution (contd.)



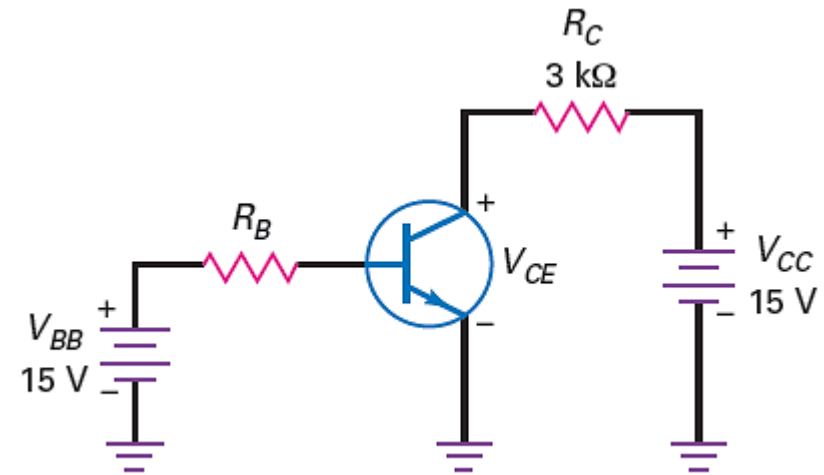
Load Line and Q point Plotting

- Draw the load line and plot the Q point for the following circuit.
- First of all we will find the end points of the load line, which are $I_{C(sat)} = 5\text{mA}$ and $V_{CE(cutt\ off)} = 15\text{V}$

$$I_B = \frac{15\text{ V}}{500\text{ k}\Omega} = 30\text{ }\mu\text{A}$$

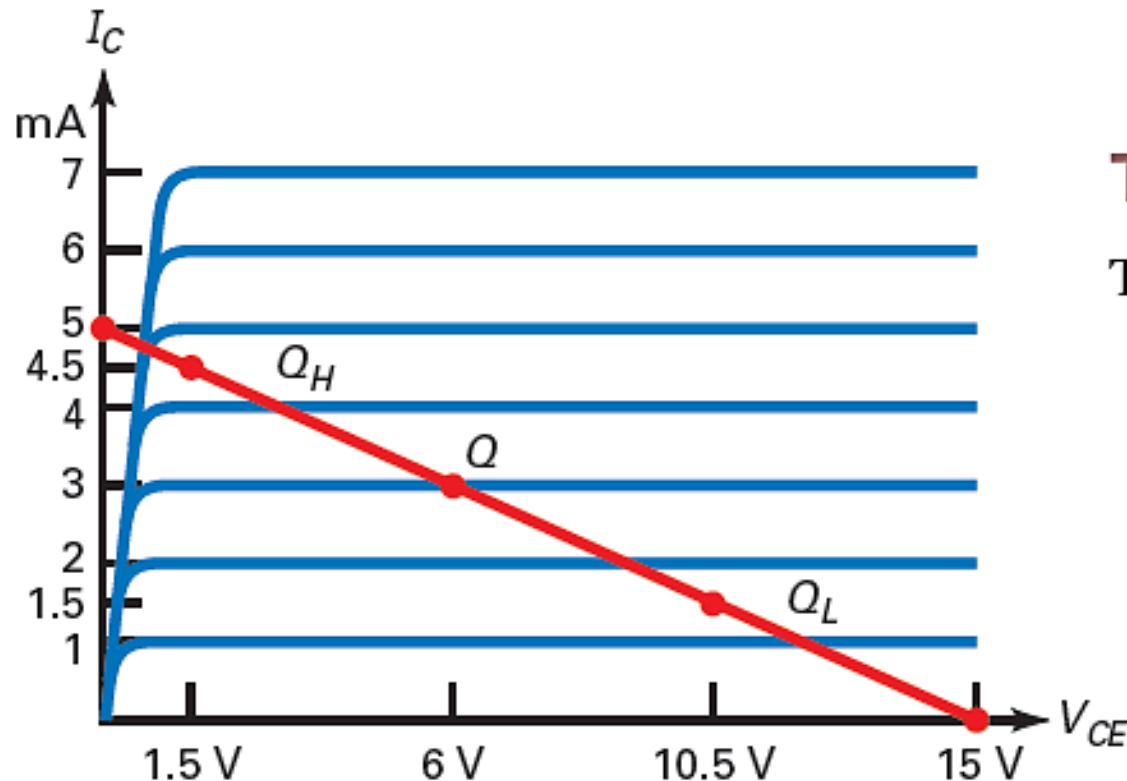
$$I_C = 100(30\text{ }\mu\text{A}) = 3\text{ mA}$$

$$V_{CE} = 15\text{ V} - (3\text{ mA})(3\text{ k}\Omega) = 6\text{ V}$$



Load Line and Q point Plotting (Contd.)

- Now if we change the value of beta
- If $\beta_{dc} = 50$ then $I_c = 1.5\text{mA}$ and $V_{CE} = 10.5\text{V}$
- If $\beta_{dc} = 150$ then $I_c = 4.5\text{mA}$ and $V_{CE} = 1.5\text{V}$



The Formulas

The formulas for calculating the Q point are as follows:

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

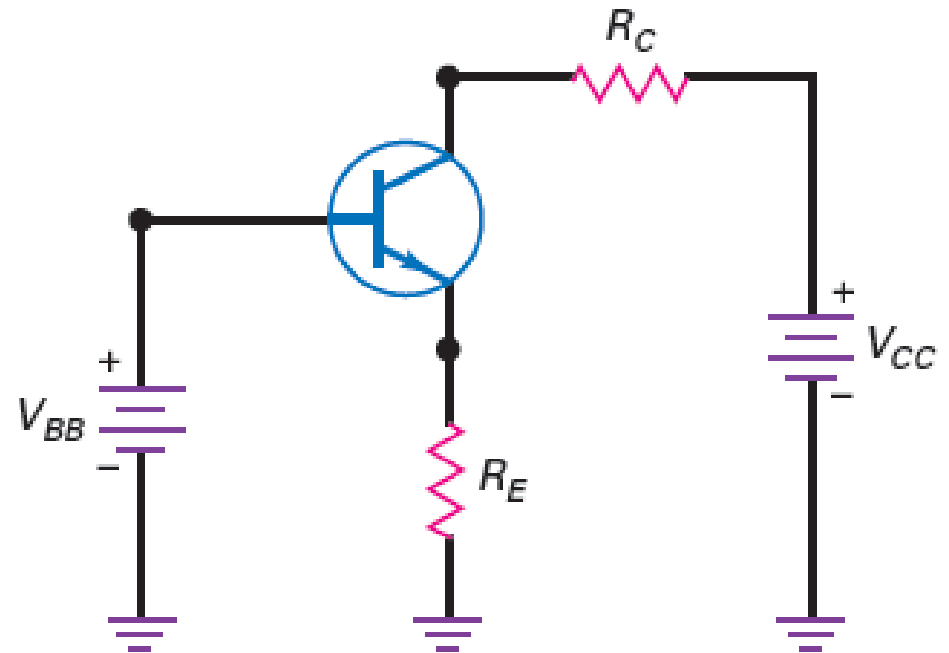
$$I_C = \beta_{dc} I_B$$

$$V_{CE} = V_{CC} - I_C R_C$$

Emitter Bias Configuration

- In previous configuration, the Q point was not stable on the load line so the goal is to make it stable. The solution is **Emitter Bias Configuration**
- The resistor from Base is shifted to the Emitter and Emitter voltage can be find

$$V_E = V_{BB} - V_{BE}$$



Emitter Bias Configuration (contd.)

- If we want to find the Q point of the Emitter Bias configuration then

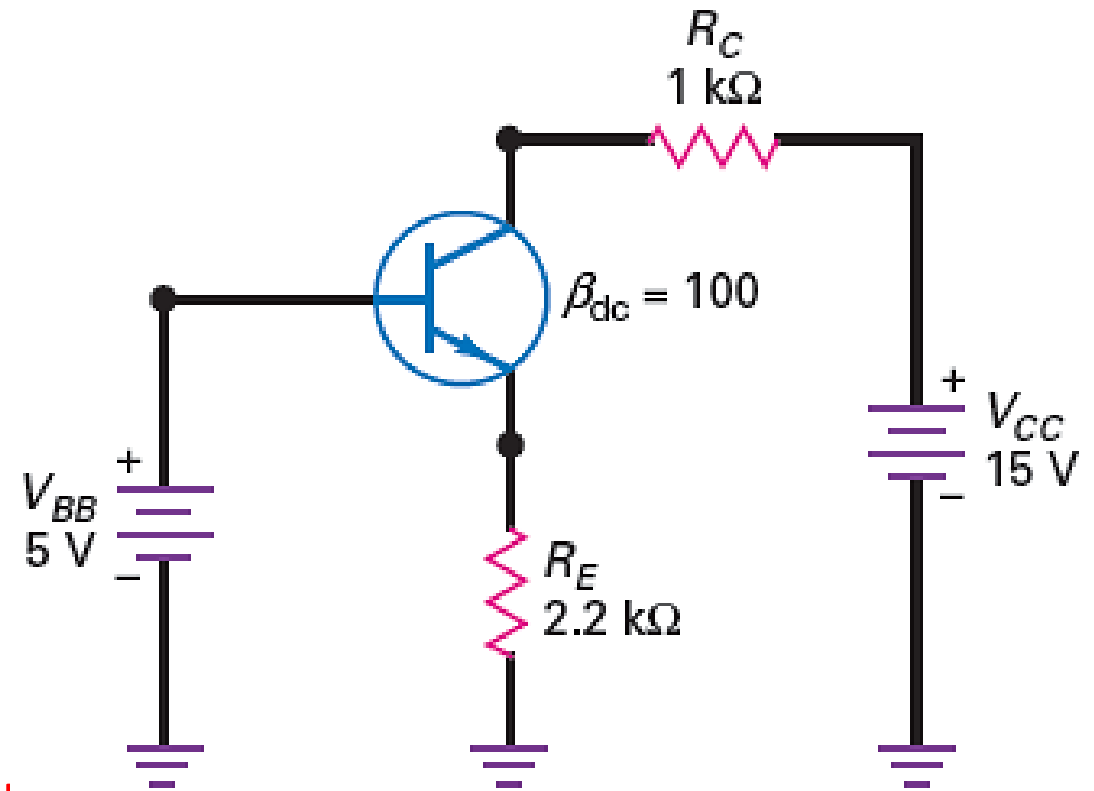
$$V_E = 5 \text{ V} - 0.7 \text{ V} = 4.3 \text{ V}$$

$$I_E = \frac{4.3 \text{ V}}{2.2 \text{ k}\Omega} = 1.95 \text{ mA}$$

$$V_C = 15 \text{ V} - (1.95 \text{ mA})(1 \text{ k}\Omega) = 13.1 \text{ V}$$

$$V_{CE} = 13.1 \text{ V} - 4.3 \text{ V} = 8.8 \text{ V}$$

- So the Q point will be (8.8V, 1.95mA)



Emitter Bias Configuration (contd.)

- The Q point for the Emitter bias circuit is immune to the changes in current gain i.e. β because there is no β involved in the calculation of Q point:
 - Get the Emitter Voltage
 - Calculate the Emitter Current
 - Find the Collector Voltage
 - Subtract the Emitter from the Collector voltage to get V_{CE}
- The current gain has a minor effect on collector current

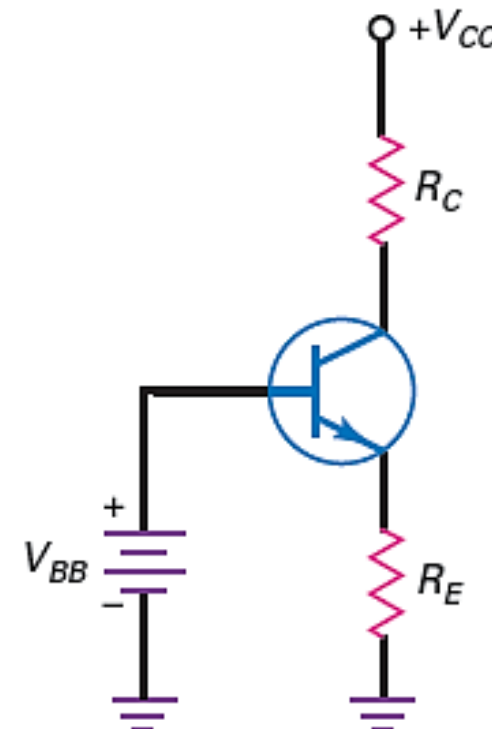
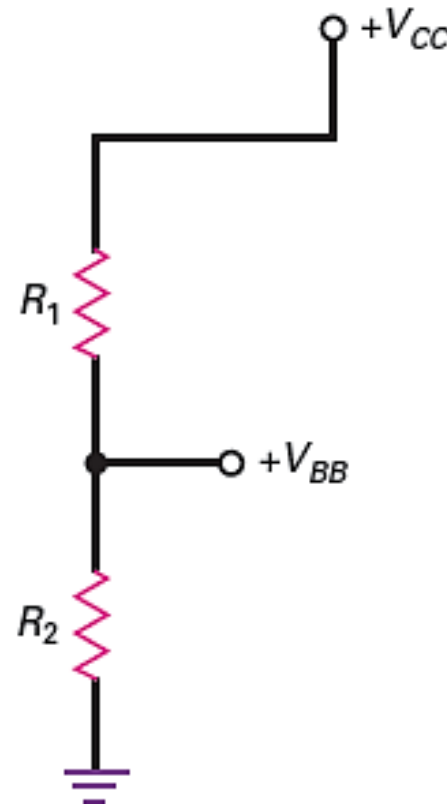
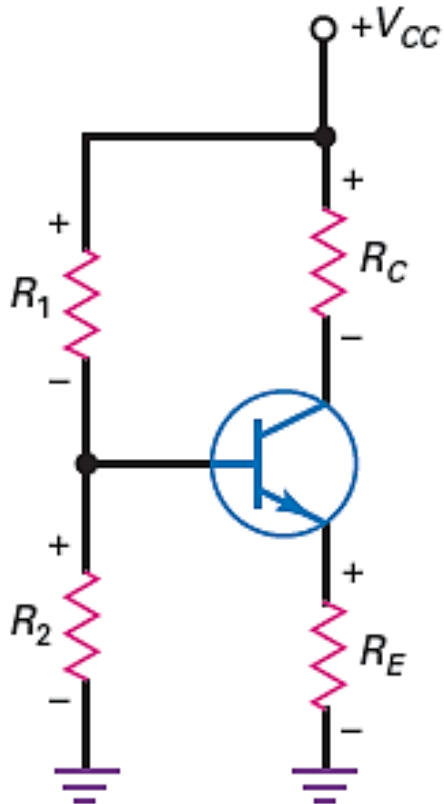
$$I_E = I_C + I_B$$

$$I_E = I_C + \frac{I_C}{\beta_{dc}}$$

$$I_C = \frac{\beta_{dc}}{\beta_{dc} + 1} I_E$$

Voltage Divider Bias Configuration

- It is most widely used configuration
- At base a voltage divider network, hence called voltage divider bias



Voltage Divider Bias Configuration (contd.)

- Analysis for the Q point can be as follows

$$V_{BB} = \frac{R_2}{R_1 + R_2} V_{CC}$$

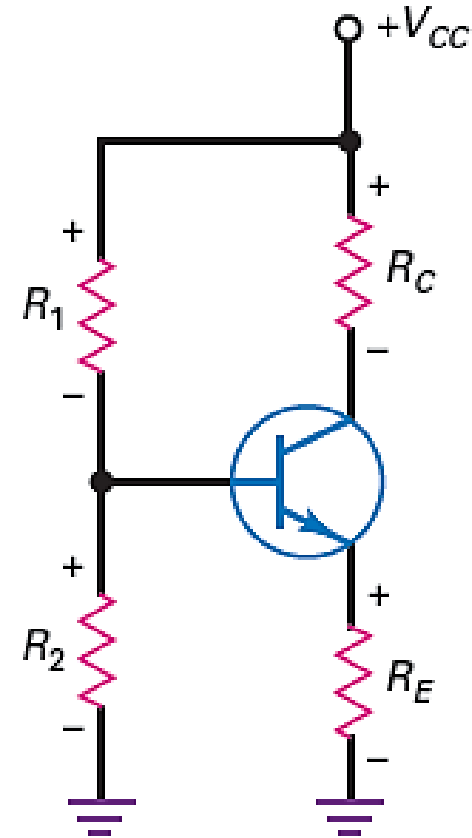
$$V_E = V_{BB} - V_{BE}$$

$$I_E = \frac{V_E}{R_E}$$

$$I_C \approx I_E$$

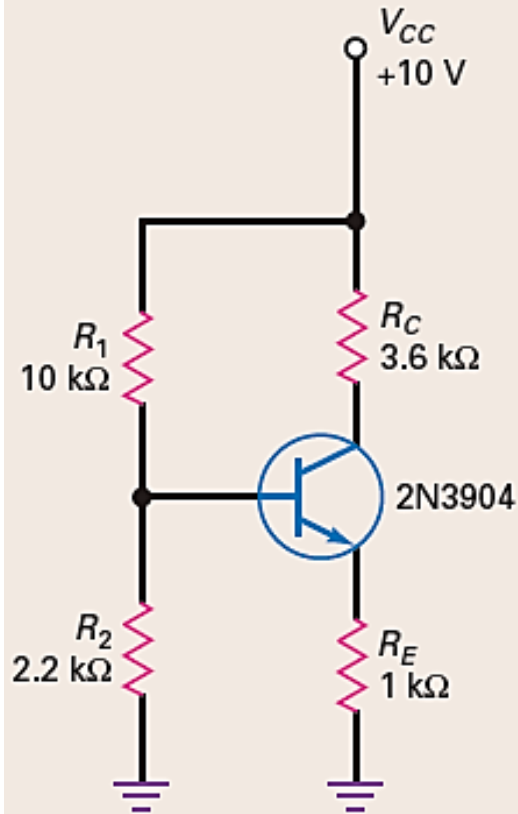
$$V_C = V_{CC} - I_C R_C$$

$$V_{CE} = V_C - V_E$$



Voltage Divider Bias Configuration (contd.)

- Find the Q point for the circuit given



SOLUTION The voltage divider produces an unloaded output voltage of:

$$V_{BB} = \frac{2.2\text{ k}\Omega}{10\text{ k}\Omega + 2.2\text{ k}\Omega} 10\text{ V} = 1.8\text{ V}$$

Subtract 0.7 V from this to get:

$$V_E = 1.8\text{ V} - 0.7\text{ V} = 1.1\text{ V}$$

The emitter current is:

$$I_E = \frac{1.1\text{ V}}{1\text{ k}\Omega} = 1.1\text{ mA}$$

Since the collector current almost equals the emitter current, we can calculate the collector-to-ground voltage like this:

$$V_C = 10\text{ V} - (1.1\text{ mA})(3.6\text{ k}\Omega) = 6.04\text{ V}$$

The collector-emitter voltage is:

$$V_{CE} = 6.04 - 1.1\text{ V} = 4.94\text{ V}$$

Here is an important point: The calculations in this preliminary analysis do not depend on changes in the transistor, the collector current, or the temperature. This is why the Q point of this circuit is stable, almost rock-solid.